BIS629: Homework 1 - Solution

1a. Prove that the "classic" test for the difference between two sample means, $H_0 : \mu_1 = \mu_2$, is the same as the likelihood-based Wald test for this same hypothesis, when $Y_j \sim N(\beta_0 + \beta^* X_j, \sigma_1^2 + X_j(\sigma_2^2 - \sigma_1^2))$ for $j = 1, ..., n_1 + n_2$, and the sample variances are known. Be sure to use the likelihood given here, parameterized by $(\beta_0, \beta^*, \sigma_1^2, \sigma_2^2)$ as the basis for this derivation. Here, X is a binary variable with value 1 if participant j is in the intervention group(group 2), and 0 otherwise, and $\sum_{j=1}^{n_1+n_2} X_j = n_2$. Begin by specifying H_0 in terms of β_0 and β^*.

1b. Now, show under what conditions, if any, the likelihood-based Wald test for these data continues to have the same form as the "classic test" when the sample variances are not known.

Note: When the sample variances are known, the "classic test" has form

$$Z^2 = \frac{(\hat{\mu}_1 - \hat{\mu}_2)^2}{\sigma_1^2/n_1 + \sigma_2^2/n_2},$$

and when the sample variances are not known, the "classic test" has form

$$Z^2 = \frac{(\hat{\mu}_1 - \hat{\mu}_2)^2}{\hat{\sigma}_1^2/n_1 + \hat{\sigma}_2^2/n_2},$$

where $\hat{\sigma}_1^2$ is the standard sample variance estimator, $\hat{\sigma}_1^2 = \frac{\sum_{j=1}^{n_1+n_2} (1-X_j)(Y_j - \hat{\mu}_1)^2}{n_1 - 1}$, $\hat{\mu}_1 = \frac{\sum_{j=1}^{n_1+n_2} (1-X_j) Y_j}{n_1}$, and likewise for $\hat{\sigma}_2^2$.

Solution:

1a. Under the given model, when $X_j = 0$ (group 1), we have $Y_j \sim N(\mu_1, \sigma_1^2)$, where $\mu_1 = \beta_0$. Similarly, when $X_j = 1$ (group 2), we have $Y_j \sim N(\mu_2, \sigma_2^2)$, where $\mu_2 = \beta_0 + \beta^*$. Therefore, the hypothesis

$$H_0 : \mu_1 = \mu_2 \quad \text{is equivalent to} \quad H_0 : \beta^* = 0.$$

The likelihood function for the given model is

$$\begin{aligned} L(\beta_0, \beta^*, \sigma_1^2, \sigma_2^2; \boldsymbol{y}) &= \prod_{j=1}^{n_1+n_2} f(Y_j \mid X_j; \beta_0, \beta^*, \sigma_1^2, \sigma_2^2) \\ &= \prod_{j:X_j=0} \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left\{-\frac{(Y_j - \beta_0)^2}{2\sigma_1^2}\right\} \\ &\quad \times \prod_{j:X_j=1} \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left\{-\frac{(Y_j - \beta_0 - \beta^*)^2}{2\sigma_2^2}\right\}. \end{aligned}$$

Then, the log-likelihood is

$$\begin{aligned} \ell(\beta_0, \beta^*) = &-\frac{n_1}{2}\log(2\pi\sigma_1^2) - \frac{1}{2\sigma_1^2}\sum_{j:X_j=0}(Y_j - \beta_0)^2 \\ &- \frac{n_2}{2}\log(2\pi\sigma_2^2) - \frac{1}{2\sigma_2^2}\sum_{j:X_j=1}(Y_j - \beta_0 - \beta^*)^2. \end{aligned}$$

We take derivatives of $\ell(\beta_0, \beta^*)$ w.r.t β_0 and β^*,

$$\frac{\partial \ell}{\partial \beta_0} = \frac{1}{\sigma_1^2}\sum_{j:X_j=0}(Y_j - \beta_0) + \frac{1}{\sigma_2^2}\sum_{j:X_j=1}(Y_j - \beta_0 - \beta^*), \tag{1}$$

$$\frac{\partial \ell}{\partial \beta^*} = \frac{1}{\sigma_2^2}\sum_{j:X_j=1}(Y_j - \beta_0 - \beta^*). \tag{2}$$

Define $\frac{1}{n_1}\sum_{j:X_j=0} Y_j = \bar{Y}_1 = \hat{\mu}_1$ and $\frac{1}{n_2}\sum_{j:X_j=1} Y_j = \bar{Y}_2 = \hat{\mu}_2$, then we further set the

equations (1) and (2) equal to zeros,

$$\frac{n_1}{\sigma_1^2}(\hat{\mu}_1 - \hat{\beta}_0) + \frac{n_2}{\sigma_2^2}(\hat{\mu}_2 - \hat{\beta}_0 - \hat{\beta}^*) = 0, \tag{3}$$

$$\hat{\mu}_2 - \hat{\beta}_0 - \hat{\beta}^* = 0. \tag{4}$$

From the equation (4), we have $\hat{\beta}^* = \hat{\mu}_2 - \hat{\beta}_0$, which can be plugged into the equation (3):

$$\frac{n_1}{\sigma_1^2}(\hat{\mu}_1 - \hat{\beta}_0) = 0 \quad \Rightarrow \quad \hat{\beta}_0 = \hat{\mu}_1.$$

Therefore, $\hat{\beta}^* = \hat{\mu}_2 - \hat{\mu}_1$. We take the second derivatives of $\ell(\beta_0, \beta^*)$ w.r.t β_0 and β^*,

$$\begin{aligned}
\frac{\partial^2 \ell}{\partial \beta_0^2} &= -\frac{n_1}{\sigma_1^2} - \frac{n_2}{\sigma_2^2}, \\
\frac{\partial^2 \ell}{\partial \beta_0 \partial \beta^*} &= -\frac{n_2}{\sigma_2^2}, \\
\frac{\partial^2 \ell}{\partial (\beta^*)^2} &= -\frac{n_2}{\sigma_2^2}.
\end{aligned}$$

Note that the second derivatives above do not depend on Y, so the observed information and the expected Fisher information coincide here. Take the negative expectation with respect to $Y|X$ for each second derivative, we get the same values:

$$\begin{aligned}
-\mathbb{E}_{Y|X}\left[\frac{\partial^2 \ell}{\partial \beta_0^2}\right] &= \frac{n_1}{\sigma_1^2} + \frac{n_2}{\sigma_2^2}, \\
-\mathbb{E}_{Y|X}\left[\frac{\partial^2 \ell}{\partial \beta_0 \partial \beta^*}\right] &= \frac{n_2}{\sigma_2^2}, \\
-\mathbb{E}_{Y|X}\left[\frac{\partial^2 \ell}{\partial (\beta^*)^2}\right] &= \frac{n_2}{\sigma_2^2}.
\end{aligned}$$

Given that the Fisher information matrix is

$$I = -\mathbb{E}_{Y|X}\left[\begin{pmatrix} \frac{\partial^2 \ell}{\partial \beta_0^2} & \frac{\partial^2 \ell}{\partial \beta_0 \partial \beta^*} \\ \frac{\partial^2 \ell}{\partial \beta^* \partial \beta_0} & \frac{\partial^2 \ell}{\partial (\beta^*)^2} \end{pmatrix}\right] = \begin{pmatrix} \frac{n_1}{\sigma_1^2} + \frac{n_2}{\sigma_2^2} & \frac{n_2}{\sigma_2^2} \\ \frac{n_2}{\sigma_2^2} & \frac{n_2}{\sigma_2^2} \end{pmatrix}, \tag{5}$$

where the variance of $\hat{\beta}^*$ is the $(2,2)$-th element of I^{-1}. Using the formula for the inverse of a 2×2 matrix,

$$\begin{aligned} \mathrm{Var}(\hat{\beta}^*) &= [I^{-1}]_{22} \\ &= \frac{\frac{n_1}{\sigma_1^2} + \frac{n_2}{\sigma_2^2}}{(\frac{n_1}{\sigma_1^2} + \frac{n_2}{\sigma_2^2})\frac{n_2}{\sigma_2^2} - (-\frac{n_2}{\sigma_2^2})^2} \\ &= \frac{\frac{n_1}{\sigma_1^2} + \frac{n_2}{\sigma_2^2}}{\frac{n_1 n_2}{\sigma_1^2 \sigma_2^2}} = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}. \end{aligned}$$

As a result, the Wald test statistic for $H_0 : \beta^* = 0$ is

$$Z_{\text{Wald}} = \frac{\hat{\beta}^* - 0}{\sqrt{\mathrm{Var}(\hat{\beta}^*)}} = \frac{\hat{\mu}_2 - \hat{\mu}_1}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}} = Z_{\text{classic}}.$$

Therefore, the classic test statistic and the likelihood-based Wald test statistic are identical.

1b. When the sample variances are unknown, we need to estimate $(\beta_0, \beta^*, \sigma_1^2, \sigma_2^2)$ from the likelihood. The log-likelihood is

$$\begin{aligned} \ell(\beta_0, \beta^*, \sigma_1^2, \sigma_2^2) = &-\frac{n_1}{2}\log(2\pi\sigma_1^2) - \frac{1}{2\sigma_1^2}\sum_{j:X_j=0}(Y_j - \beta_0)^2 \\ &- \frac{n_2}{2}\log(2\pi\sigma_2^2) - \frac{1}{2\sigma_2^2}\sum_{j:X_j=1}(Y_j - \beta_0 - \beta^*)^2. \end{aligned}$$

We take derivatives w.r.t all four parameters. From the problem 1a, we already have

$\hat{\beta}_0 = \hat{\mu}_1$ and $\hat{\beta}^* = \hat{\mu}_2 - \hat{\mu}_1$. For the variance parameters, we have

$$\frac{\partial \ell}{\partial \sigma_1^2} = -\frac{n_1}{2\sigma_1^2} + \frac{1}{2(\sigma_1^2)^2} \sum_{j:X_j=0} (Y_j - \beta_0)^2, \tag{6}$$

$$\frac{\partial \ell}{\partial \sigma_2^2} = -\frac{n_2}{2\sigma_2^2} + \frac{1}{2(\sigma_2^2)^2} \sum_{j:X_j=1} (Y_j - \beta_0 - \beta^*)^2. \tag{7}$$

We further set equations (6) and (7) equal to zeros,

$$\hat{\sigma}_{1,\text{MLE}}^2 = \frac{1}{n_1} \sum_{j:X_j=0} (Y_j - \hat{\beta}_0)^2 = \frac{1}{n_1} \sum_{j=1}^{n_1+n_2} (1 - X_j)(Y_j - \hat{\mu}_1)^2,$$

$$\hat{\sigma}_{2,\text{MLE}}^2 = \frac{1}{n_2} \sum_{j:X_j=1} (Y_j - \hat{\beta}_0 - \hat{\beta}^*)^2 = \frac{1}{n_2} \sum_{j=1}^{n_1+n_2} X_j (Y_j - \hat{\mu}_2)^2.$$

Note that the Fisher information matrix for $(\beta_0, \beta^*, \sigma_1^2, \sigma_2^2)$ is block diagonal because of the orthogonality of the mean and variance parameters in the normal distribution. To show that, we have

$$\frac{\partial^2 \ell}{\partial \sigma_1^2 \partial \beta_0} = -\frac{1}{(\sigma_1^2)^2} \sum_{j:X_j=0} (Y_j - \beta_0),$$

$$\frac{\partial^2 \ell}{\partial \sigma_1^2 \partial \beta^*} = 0,$$

$$\frac{\partial^2 \ell}{\partial \sigma_2^2 \partial \beta_0} = -\frac{1}{(\sigma_2^2)^2} \sum_{j:X_j=1} (Y_j - \beta_0 - \beta^*),$$

$$\frac{\partial^2 \ell}{\partial \sigma_2^2 \partial \beta^*} = -\frac{1}{(\sigma_2^2)^2} \sum_{j:X_j=1} (Y_j - \beta_0 - \beta^*).$$

The expectations of the partial derivatives above are zero:

$$-\mathbb{E}_{Y|X}\left[\frac{\partial^2 \ell}{\partial \sigma_1^2 \partial \beta_0}\right] = -\mathbb{E}_{Y|X}\left[\frac{\partial^2 \ell}{\partial \sigma_1^2 \partial \beta^*}\right] = -\mathbb{E}_{Y|X}\left[\frac{\partial^2 \ell}{\partial \sigma_2^2 \partial \beta_0}\right] = -\mathbb{E}_{Y|X}\left[\frac{\partial^2 \ell}{\partial \sigma_2^2 \partial \beta^*}\right] = 0$$

Therefore, the expected Fisher Information matrix for $(\beta_0, \beta^*, \sigma_1^2, \sigma_2^2)$ can be written as:

$$I = \begin{pmatrix} I_\beta & \mathbf{0}_{2\times 2} \\ \mathbf{0}_{2\times 2} & I_\sigma \end{pmatrix}.$$

Note that the second derivatives respect to σ_1^2 and σ_2^2 are not calculated here because the expected cross-information between the mean and variance parameters is zero, as shown above. Therefore, the Fisher information matrix is block diagonal, and the covariance of the mean parameter estimates $\hat{\beta}_0$ and $\hat{\beta}^*$ can be obtained by inverting the mean parameter block I_β alone. The variance parameter second derivatives are not needed for that calculation. Hence,

$$I^{-1} = \begin{pmatrix} I_\beta^{-1} & \mathbf{0}_{2\times 2} \\ \mathbf{0}_{2\times 2} & I_\sigma^{-1} \end{pmatrix}.$$

Here, I_β is a 2×2 matrix for (β_0, β^*), which is given by equation (5). As shown in 1a,

$$\mathrm{Var}(\hat{\beta}^*) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}.$$

When σ_1^2 and σ_2^2 are unknown, the plug-in estimator of $\mathrm{Var}(\hat{\beta}^*)$ is

$$\widehat{\mathrm{Var}}(\hat{\beta}^*) = \frac{\hat{\sigma}_{1,\mathrm{MLE}}^2}{n_1} + \frac{\hat{\sigma}_{2,\mathrm{MLE}}^2}{n_2}.$$

Therefore, the Wald test statistic is

$$Z_{\mathrm{Wald}}^2 = \frac{(\hat{\beta}^*)^2}{\widehat{\mathrm{Var}}(\hat{\beta}^*)} = \frac{(\hat{\mu}_2 - \hat{\mu}_1)^2}{\hat{\sigma}_{1,\mathrm{MLE}}^2/n_1 + \hat{\sigma}_{2,\mathrm{MLE}}^2/n_2}.$$

Recall that the "classic" test can be written as

$$\hat{\sigma}^2_{1,\text{classic}} = \frac{1}{n_1 - 1} \sum_{j=1}^{n_1+n_2} (1 - X_j)(Y_j - \hat{\mu}_1)^2,$$
$$\hat{\sigma}^2_{2,\text{classic}} = \frac{1}{n_2 - 1} \sum_{j=1}^{n_1+n_2} X_j(Y_j - \hat{\mu}_2)^2.$$

Therefore, the two tests have the same form if

$$\frac{\hat{\sigma}^2_{1,\text{MLE}}}{n_1} = \frac{\hat{\sigma}^2_{1,\text{classic}}}{n_1} \quad \text{and} \quad \frac{\hat{\sigma}^2_{2,\text{MLE}}}{n_2} = \frac{\hat{\sigma}^2_{2,\text{classic}}}{n_2}.$$

Since $\hat{\sigma}^2_{1,\text{MLE}} = \frac{n_1-1}{n_1}\hat{\sigma}^2_{1,\text{classic}}$ and $\hat{\sigma}^2_{2,\text{MLE}} = \frac{n_2-1}{n_2}\hat{\sigma}^2_{2,\text{classic}}$, the tests are equivalent only when $\frac{n_1-1}{n_1} \approx 1$ and $\frac{n_2-1}{n_2} \approx 1$, which occurs when both n_1 and n_2 are large. The two tests are asymptotically equivalent and the difference will usually be negligible when both n_1 and n_2 are large.

2. Show that the "classic" χ^2 test for the difference between two proportions, $H_0 : p_1 = p_2$, is the likelihood-based Wald test when $Y_j \sim Bern(\beta_0 + \beta^* X_j)$ for $j = 1, ..., n_1 + n_2$. Here, X is a binary variable with value 1 if participant j is in the intervention group, and 0 otherwise, and $\sum_{j=1}^{n_1+n_2} X_j = n_2$. Begin by specifying H_0 in terms of β_0 and β^*.

Note: One way of writing the "classic" χ^2 test is

$$Z^2 = \frac{(\hat{p}_1 - \hat{p}_2)^2}{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}},$$

where $\hat{p}_1 = \frac{\sum_{j=1}^{n_1+n_2}(1-X_j)Y_j}{n_1}$ and $\hat{p}_2 = \frac{\sum_{j=1}^{n_1+n_2} X_j Y_j}{n_2}$.

Solution:

Under the given model, when $X_j = 0$ (group 1), we have $Y_j \sim \text{Bern}(p_1)$, where $p_1 = \beta_0$. Similarly, when $X_j = 1$ (group 2), we have $Y_j \sim \text{Bern}(p_2)$, where $p_2 = \beta_0 + \beta^*$. Therefore, the hypothesis

$$H_0 : p_1 = p_2 \quad \text{is equivalent to} \quad H_0 : \beta^* = 0.$$

The likelihood function for the Bernoulli model is

$$\begin{aligned} L(\beta_0, \beta^*; \boldsymbol{y}) &= \prod_{j=1}^{n_1+n_2} (\beta_0 + \beta^* X_j)^{Y_j} (1 - \beta_0 - \beta^* X_j)^{1-Y_j} \\ &= \prod_{j:X_j=0} \beta_0^{Y_j} (1-\beta_0)^{1-Y_j} \times \prod_{j:X_j=1} (\beta_0 + \beta^*)^{Y_j} (1 - \beta_0 - \beta^*)^{1-Y_j}. \end{aligned}$$

Then, the log-likelihood is

$$\begin{aligned} \ell(\beta_0, \beta^*) = &\sum_{j:X_j=0} [Y_j \log(\beta_0) + (1 - Y_j) \log(1 - \beta_0)] \\ &+ \sum_{j:X_j=1} [Y_j \log(\beta_0 + \beta^*) + (1 - Y_j) \log(1 - \beta_0 - \beta^*)]. \end{aligned}$$

We take derivatives of $\ell(\beta_0, \beta^*)$ w.r.t β_0 and β^*:

$$\frac{\partial \ell}{\partial \beta_0} = \sum_{j:X_j=0} \left(\frac{Y_j}{\beta_0} - \frac{1-Y_j}{1-\beta_0} \right) + \sum_{j:X_j=1} \left(\frac{Y_j}{\beta_0+\beta^*} - \frac{1-Y_j}{1-\beta_0-\beta^*} \right), \tag{8}$$

$$\frac{\partial \ell}{\partial \beta^*} = \sum_{j:X_j=1} \left(\frac{Y_j}{\beta_0+\beta^*} - \frac{1-Y_j}{1-\beta_0-\beta^*} \right). \tag{9}$$

We further set the equation (9) equal to zero,

$$\sum_{j:X_j=1} Y_j(1-\beta_0-\beta^*) = \sum_{j:X_j=1} (1-Y_j)(\beta_0+\beta^*)$$

$$n_2\hat{p}_2(1-\hat{\beta}_0-\hat{\beta}^*) = n_2(1-\hat{p}_2)(\hat{\beta}_0+\hat{\beta}^*),$$

where $\hat{p}_2 = n_2^{-1}\sum_{j=1}^{n_1+n_2} X_j Y_j$. This implies that $\hat{\beta}_0 + \hat{\beta}^* = \hat{p}_2$. Similarly, setting the equation (8) equals to zero and using the result from equation (9), we have

$$\sum_{j:X_j=0} \left[\frac{Y_j}{\hat{\beta}_0} - \frac{1-Y_j}{1-\hat{\beta}_0} \right] = 0$$

$$n_1\hat{p}_1(1-\hat{\beta}_0) = n_1(1-\hat{p}_1)\hat{\beta}_0,$$

where $\hat{p}_1 = n_1^{-1}\sum_{j=1}^{n_1+n_2}(1-X_j)Y_j$. This implies that $\hat{\beta}_0 = \hat{p}_1$. Therefore, $\hat{\beta}^* = \hat{p}_2 - \hat{p}_1$. Given that the Fisher information matrix is

$$I = -\mathbb{E}_{Y|X}\left[\begin{pmatrix} \frac{\partial^2 \ell}{\partial \beta_0^2} & \frac{\partial^2 \ell}{\partial \beta_0 \partial \beta^*} \\ \frac{\partial^2 \ell}{\partial \beta^* \partial \beta_0} & \frac{\partial^2 \ell}{\partial (\beta^*)^2} \end{pmatrix} \right].$$

We proceed to computing the second derivatives

$$\begin{aligned}
\frac{\partial^2 \ell}{\partial \beta_0^2} &= \sum_{j:X_j=0} \left(-\frac{Y_j}{\beta_0^2} - \frac{1-Y_j}{(1-\beta_0)^2} \right) + \sum_{j:X_j=1} \left(-\frac{Y_j}{(\beta_0+\beta^*)^2} - \frac{1-Y_j}{(1-\beta_0-\beta^*)^2} \right), \\
\frac{\partial^2 \ell}{\partial \beta_0 \partial \beta^*} &= \sum_{j:X_j=1} \left(-\frac{Y_j}{(\beta_0+\beta^*)^2} - \frac{1-Y_j}{(1-\beta_0-\beta^*)^2} \right), \\
\frac{\partial^2 \ell}{\partial (\beta^*)^2} &= \sum_{j:X_j=1} \left(-\frac{Y_j}{(\beta_0+\beta^*)^2} - \frac{1-Y_j}{(1-\beta_0-\beta^*)^2} \right)
\end{aligned}$$

Since $Y_j \sim Bern(\beta_0 + \beta^* X_j)$, $E[Y_j|X_j] = \beta_0 + \beta^* X_j$. Then take the negative expectation with respect to $Y|X$ for each second derivative:

$$\begin{aligned}
-\mathbb{E}_{Y|X}\left[\frac{\partial^2 \ell}{\partial \beta_0^2}\right] &= \sum_{j:X_j=0} \left(\frac{E[Y_j|X_j=0]}{\beta_0^2} + \frac{1-E[Y_j|X_j=0]}{(1-\beta_0)^2} \right) \\
&\quad + \sum_{j:X_j=1} \left(\frac{E[Y_j|X_j=1]}{(\beta_0+\beta^*)^2} + \frac{1-E[Y_j|X_j=1]}{(1-\beta_0-\beta^*)^2} \right) \\
&= \sum_{j=1}^{n_1} \left(\frac{\beta_0}{\beta_0^2} + \frac{1-\beta_0}{(1-\beta_0)^2} \right) + \sum_{j=n_1+1}^{n_1+n_2} \left(\frac{\beta_0+\beta^*}{(\beta_0+\beta^*)^2} + \frac{1-(\beta_0+\beta^*)}{(1-\beta_0-\beta^*)^2} \right) \\
&= \frac{n_1}{\beta_0(1-\beta_0)} + \frac{n_2}{(\beta_0+\beta^*)(1-\beta_0-\beta^*)}, \\
-\mathbb{E}_{Y|X}\left[\frac{\partial^2 \ell}{\partial \beta_0 \partial \beta^*}\right] &= \sum_{j:X_j=1} \left(\frac{E[Y_j|X_j=1]}{(\beta_0+\beta^*)^2} + \frac{1-E[Y_j|X_j=1]}{(1-\beta_0-\beta^*)^2} \right) \\
&= \sum_{j=n_1+1}^{n_1+n_2} \left(\frac{\beta_0+\beta^*}{(\beta_0+\beta^*)^2} + \frac{1-(\beta_0+\beta^*)}{(1-\beta_0-\beta^*)^2} \right) \\
&= \frac{n_2}{(\beta_0+\beta^*)(1-\beta_0-\beta^*)}, \\
-\mathbb{E}_{Y|X}\left[\frac{\partial^2 \ell}{\partial (\beta^*)^2}\right] &= \sum_{j:X_j=1} \left(\frac{E[Y_j|X_j=1]}{(\beta_0+\beta^*)^2} + \frac{1-E[Y_j|X_j=1]}{(1-\beta_0-\beta^*)^2} \right) \\
&= \sum_{j=n_1+1}^{n_1+n_2} \left(\frac{\beta_0+\beta^*}{(\beta_0+\beta^*)^2} + \frac{1-(\beta_0+\beta^*)}{(1-\beta_0-\beta^*)^2} \right) \\
&= \frac{n_2}{(\beta_0+\beta^*)(1-\beta_0-\beta^*)}.
\end{aligned}$$

We evaluate at the MLEs $\hat{\beta}_0 = \hat{p}_1$ and $\hat{\beta}_0 + \hat{\beta}^* = \hat{p}_2$:

$$\hat{I} = \begin{pmatrix} \frac{n_1}{\hat{p}_1(1-\hat{p}_1)} + \frac{n_2}{\hat{p}_2(1-\hat{p}_2)} & \frac{n_2}{\hat{p}_2(1-\hat{p}_2)} \\ \frac{n_2}{\hat{p}_2(1-\hat{p}_2)} & \frac{n_2}{\hat{p}_2(1-\hat{p}_2)} \end{pmatrix}.$$

The variance of $\hat{\beta}^*$ is the $(2,2)$-th element of $\hat{I}^{-1}$:

$$\begin{aligned} \widehat{\mathrm{Var}}(\hat{\beta}^*) &= [\hat{I}^{-1}]_{22} \\ &= \frac{\frac{n_1}{\hat{p}_1(1-\hat{p}_1)} + \frac{n_2}{\hat{p}_2(1-\hat{p}_2)}}{(\frac{n_1}{\hat{p}_1(1-\hat{p}_1)} + \frac{n_2}{\hat{p}_2(1-\hat{p}_2)})\frac{n_2}{\hat{p}_2(1-\hat{p}_2)} - (-\frac{n_2}{\hat{p}_2(1-\hat{p}_2)})^2} \\ &= \frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}. \end{aligned}$$

As a result, the Wald test statistic for $H_0 : \beta^* = 0$ is

$$Z^2_{\text{Wald}} = \frac{(\hat{\beta}^*)^2}{\widehat{\mathrm{Var}}(\hat{\beta}^*)} = \frac{(\hat{p}_2 - \hat{p}_1)^2}{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} = Z^2_{\text{classic}}.$$

Therefore, the classic χ^2 test statistic and the likelihood-based Wald test statistic are identical.

The following two questions are based on MaxART data (Khan et al. 2020*). The data set "MaxART HW1 2026.RData" is in the Homework 1 folder in Canvas. The outcome variable is named "Y", where 1 means CD4 counts at 18 months were $\geq$ 350 cells/mm^3, and 0 otherwise, "X" is the treatment indicator, where 1 indicates that the clinic at enrollment was randomized to early access to ART as soon as testing positive to HIV, and 0 indicating the clinic at which the participant enrolled was randomized to standard of care.

3. Test H_0: the intervention has no effect on achieving CD4 counts $\geq$ 350 cells/mm^3 at 18 months vs. H_A: the intervention has an effect on achieving CD4 counts $\geq$ 350 cells/mm^3 at 18 months, by the Wald, score and likelihood ratio tests, using the logistic regression model. a)State the hypothesis mathematically. b) Perform the Wald, Score, and Likelihood ratio tests. c) Compare the results from the 3 test statistics in a concise, well-labeled table. d) Interpret your findings concisely from a subject matter point of view.

Solution:

3a: Mathematical hypothesis

The outcome is

$$Y_i = \begin{cases} 1, & \text{CD4 count at 18 months is at least 350 cells/mm}^3, \\ 0, & \text{CD4 count at 18 months is below 350 cells/mm}^3, \end{cases}$$

and the treatment indicator is

$$X_i = \begin{cases} 1, & \text{clinic assigned to early access to ART}, \\ 0, & \text{clinic assigned to standard of care}. \end{cases}$$

* Khan, S., Spiegelman, D., Walsh, F., Mazibuko, S., Pasipamire, M., Chai, B., Reis, R., Mlambo, K., Delva, W., Khumalo, G., Zwane, M., Fleming, Y., Mafara, E., Hettema, A., Lejeune, C., Chao, A., Bärnighausen, T., & Okello, V. (2020). Early access to antiretroviral therapy versus standard of care among HIV-positive participants in Eswatini in the public health sector: the MaxART stepped-wedge randomized controlled trial. Journal of the International AIDS Society, 23(9), e25610.

For participant i, define

$$p_i = P(Y_i = 1 \mid X_i).$$

The unadjusted logistic regression model is

$$\text{logit}(p_i) = \log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_i.$$

Under this parameterization,

$$P(Y = 1 \mid X = 0) = \text{expit}(\beta_0)$$

and

$$P(Y = 1 \mid X = 1) = \text{expit}(\beta_0 + \beta_1),$$

where $\text{expit}(a) = \exp(a)/(1+\exp(a))$.

The treatment odds ratio is

$$\text{OR} = \frac{p_1/(1-p_1)}{p_0/(1-p_0)} = \exp(\beta_1).$$

Therefore, the two-sided hypotheses are:

$$H_0 : \beta_1 = 0,$$

$$H_A : \beta_1 \neq 0.$$

Equivalently,

$$H_0 : \text{OR} = 1,$$

$$H_A : \text{OR} \neq 1.$$

Table 1: Observed outcome counts by treatment group

Group	$Y = 1$	$Y = 0$	Total	Observed probability
Standard of care ($X = 0$)	25	26	51	0.49
Early ART ($X = 1$)	621	480	1,101	0.56

Fit two logistic regression models:

- Null model: $\text{logit}(p_i) = \beta_0$.
- Alternative model: $\text{logit}(p_i) = \beta_0 + \beta_1 X_i$.

The fitted parameters are summarized below.

Table 2: Design and fitted-model quantities

Quantity	Estimate
Sample size, n	1,152
Treatment proportion, $\widehat{P}(X = 1)$	0.96
Null-model intercept, $\widehat{\beta}_{0,0}$	0.24
Alternative-model intercept, $\widehat{\beta}_{0,A}$	-0.04
Treatment log odds ratio, $\widehat{\beta}_1$	0.30
Treatment odds ratio, $\exp(\widehat{\beta}_1)$	1.35
$\widehat{P}(Y = 1 \mid X = 0)$	0.49
$\widehat{P}(Y = 1 \mid X = 1)$	0.56

3b: Wald test

The Wald z statistic is

$$Z_W = \frac{\widehat{\beta}_1 - 0}{\text{SE}(\widehat{\beta}_1)},$$

and the equivalent one-degree-of-freedom chi-square statistic is

$$W = Z_W^2 = \frac{\widehat{\beta}_1^2}{\widehat{\mathrm{Var}}(\widehat{\beta}_1)}.$$

For these data,

$$\mathrm{SE}(\widehat{\beta}_1) = 0.29, \qquad Z_W = \frac{0.2968}{0.2866} = 1.04,$$

so

$$W = (1.0354)^2 = 1.07.$$

Under H_0, W is approximately χ_1^2, giving

$$p = P(\chi_1^2 \geq 1.0720) = 0.30.$$

Because $p > 0.05$, the Wald test does not reject H_0.

3b: Score test

Let

$$\widehat{p}_{\text{null}} = \mathrm{expit}(\widehat{\beta}_{0,0}).$$

The score for β_1, evaluated under H_0, is

$$U_{\beta_1} = \sum_{i=1}^{n} X_i(Y_i - \widehat{p}_{\text{null}}).$$

Because the intercept is a nuisance parameter, the efficient information for β_1 is

$$I_{\text{eff}} = \widehat{p}_{\text{null}}(1 - \widehat{p}_{\text{null}}) \sum_{i=1}^{n} (X_i - \overline{X})^2.$$

The score statistic is

$$S = \frac{U_{\beta_1}^2}{I_{\text{eff}}}.$$

For the MaxART data,

$$S = 1.08, \qquad p = P(\chi^2_1 \geq 1.0789) = 0.30.$$

Therefore, the score test does not reject H_0.

3b: Likelihood-ratio test

The likelihood-ratio test compares the maximized likelihoods under the null and alternative models. Its statistic is

$$G^2 = 2\{\ell(\widehat{\beta}_0, \widehat{\beta}_1) - \ell(\widehat{\beta}_{0,0}, 0)\}.$$

For nested generalized linear models, the same value is obtained as the difference in model deviances:

$$G^2 = \text{Deviance(null)} - \text{Deviance(alternative)}.$$

For these data,

$$G^2 = 1.0705, \qquad p = P(\chi^2_1 \geq 1.0705) = 0.30.$$

Therefore, the likelihood-ratio test does not reject H_0.

3c: Compare results in a table

Table 3: Comparison of the three tests of $H_0 : \beta_1 = 0$

Test	Statistic	df	p-value
Wald	1.07	1	0.30
Score	1.08	1	0.30
Likelihood ratio	1.07	1	0.30

3d. Subject-matter interpretation: The data estimate higher odds of attaining CD4 ≥ 350 at 18 months under early ART, with $\widehat{\text{OR}} = 1.35$. Nevertheless, all three tests give $p \approx 0.30$. Thus, this analysis does not provide sufficient statistical evidence that early ART changes the odds of attaining the CD4 threshold. "Do not reject" does not prove that the

intervention has no effect; it means the observed data are not sufficiently precise to distinguish the estimated effect from no effect at the 5% significance level.

Question 4

4. Calculate the a) power, b) sample size and c) MDE of the odds ratio (OR) in MaxART for, as relevant, power at 80% and 90%, and significance level at 5%, and use the estimated value of $\hat{\beta}^*_{H_A}$ observed in MaxART for these calculations. Use other background design features as found in MaxART as needed, and be sure to clearly specify all parameters assumed for each of the 3 calculations, using all 3 test statistics. Create a concise, well-labeled table summarizing results.

Solution:

The model design inputs:

Table 4: Inputs used for power, sample-size, and MDE calculations

Input	Symbol	Value
Significance level	α	0.05
Treatment allocation	$p_x = P(X = 1)$	0.96
Control log odds under H_A	$\beta_{0\|H_A}$	-0.04
Treatment effect	β^*	0.30
Odds ratio	$\exp(\beta^*)$	1.35
Control outcome probability	p_0	0.49
Treatment outcome probability	p_1	0.56

NB: These tests are based on the logistic regression likelihood for binomial data, given in course notes (see Lecture_1 2026 slides 22 - 41)

Wald test calculations

For the Wald test, we define

$$I_j^{-1}(\beta^*) = \frac{1}{1-p_x}\frac{\{1+\exp(\beta_0)\}^2}{\exp(\beta_0)} + \frac{1}{p_x}\frac{\{1+\exp(\beta_0+\beta^*)\}^2}{\exp(\beta_0+\beta^*)}.$$

The Wald non-centrality parameter for a given power, π is

$$\lambda(\pi) = n\frac{(\beta^*)^2}{I_j^{-1}(\hat{\beta}^*)}.$$

Using the observed alternative effect gives

$$\lambda(\pi) = 1.07$$

and therefore

$$\text{power} = P\{\chi_1^2(1.0720) > 3.8415\} = 0.18.$$

For a target non-centrality parameter $\lambda(\pi)$, the required sample size is

$$n = \left\lceil \frac{\lambda(\pi)}{(\beta^*)^2/I_j^{-1}(\hat{\beta}^*)} \right\rceil.$$

where $\lambda(\pi)$ is the solution to,

$$P\left\{\chi_1^2(\lambda(\pi)) > 3.8415\right\} = \pi,$$

where 3.8415 is the critical value for a two-sided test with $\alpha = 0.05$ and 1 degree of freedom. Therefore:

$$\lambda(\pi) = \begin{cases} \lambda(\pi = 0.8) \approx 7.85, & \text{for 80\% power,} \\ \lambda(\pi = 0.9) \approx 10.50, & \text{for 90\% power.} \end{cases}$$

For 80% and 90% power respectively

$$n = 8{,}436, \qquad n = 11{,}284.$$

For the MDE, hold n, p_x, $\lambda(\pi)$ and $\beta_{0,A}$ fixed and solve

$$n\frac{(\beta^*_{\text{MDE}})^2}{I^{-1}_{\beta^*}(\beta^*_{\text{MDE}})} = \lambda(\pi).$$

Because the observed effect is positive, we choose the positive root and report $\exp(\beta_{1,\text{MDE}})$. The Wald MDEs for 80% and 90% power, respectively, are

$$\text{OR} = 2.24, \qquad \text{OR} = 2.54.$$

Score test calculations

Let

$$p_0 = \text{expit}(\beta_{0|H_A}), \qquad p_1 = \text{expit}(\beta_{0|H_A} + \beta_*).$$

The non-centrality parameters for the score test is given by;

$$\lambda(\pi) = n\frac{p_x(1-p_x)(p_1-p_0)^2}{p_x p_0(1-p_0) + (1-p_x)p_1(1-p_1)}.$$

At the observed effect,

$$\lambda(\pi) = 1.06,$$

which gives

$$\text{power} = 0.18.$$

Define the per-participant score as

$$S = \frac{p_x(1-p_x)(p_1-p_0)^2}{p_x p_0(1-p_0) + (1-p_x)p_1(1-p_1)}.$$

Then

$$n = \left\lceil \frac{\lambda(\pi)}{S} \right\rceil,$$

for 80% and 90% power respectively, gives,

$$n = 8{,}499, \qquad n = 11{,}368.$$

For the MDE, we solve for β^*, compute p_1, and find the β^*_{MDE} that solves the equation for $\lambda(\pi)$.

Taking the positive root gives;

$$\text{expit}(\beta^*_{\text{MDE}}) = OR$$

for 80% and 90% power respectively,

$$\text{OR} = 2.32, \qquad \text{OR} = 2.69.$$

Likelihood-ratio test calculations

For the likelihood-ratio approximation, we first define

$$\begin{aligned} p_0 &= \text{expit}(\beta_{0|H_A}), \\ p_1 &= \text{expit}(\beta_{0|H_A} + \beta^*), \\ p^* &= (1 - p_x)p_0 + p_x p_1, \\ \widetilde{\beta} &= \log\left(\frac{p^*}{1 - p^*}\right). \end{aligned}$$

The log likelihood is given by

$$\Delta^* = 2\left[p_x\left\{p_1(\beta_{0|H_A} + \beta^* - \widetilde{\beta}_0) - \log\left(\frac{1 + e^{\beta_{0|H_A} + \beta^*}}{1 + e^{\widetilde{\beta}_0}}\right)\right\} + (1 - p_x)\left\{p_0(\beta_{0|H_A} - \widetilde{\beta}_0) - \log\left(\frac{1 + e^{\beta_{0|H_A}}}{1 + e^{\widetilde{\beta}_0}}\right)\right\}\right].$$

Also define

$$M = \frac{p_x p_1(1-p_1) + (1-p_x)p_0(1-p_0)}{p^*(1-p^*)}.$$

For the observed alternative,

$$p^* = 0.56, \qquad \widetilde{\beta}_0 = 0.24,$$

$$\Delta^* = 0.000929251, \qquad M = 0.9990635.$$

The likelihood-ratio non-centrality parameter is

$$\lambda(\pi) = \Delta + 1 - M = 1.07,$$

which gives

$$\text{power} = 0.18.$$

Solving the non-centrality parameter expression for n gives

$$n = \left\lceil \frac{\lambda(\pi) - 1 + M}{\Delta^*} \right\rceil.$$

Therefore, for 80% and 90% power respectively , we have

$$n = 8{,}447, \qquad n = 11{,}299.$$

For the likelihood-ratio MDE, replacing β^* by the unknown β^*_{MDE}, we compute p_1, p^*, $\widetilde{\beta}$, Δ^*, and M, and solve

$$n\Delta^* + 1 - M = \lambda(\pi).$$

For 80% and 90% power, respectively, the MDEs are

$$\text{OR} = 2.25, \qquad \text{OR} = 2.57.$$

Concise result table

Table 5: Power, required sample size, and MDE by test

Test	Target power	Required n	MDE OR
Wald	80%	8,436	2.24
Wald	90%	11,284	2.54
Score	80%	8,499	2.32
Score	90%	11,368	2.69
Likelihood ratio	80%	8,447	2.25
Likelihood ratio	90%	11,299	2.57

The three tests produced very similar results. However, to detect the observed intervention effect, we would require approximately 8,400–8,500 participants for 80% power and 11,300–11,400 for 90% power. At the given sample size of 1152, only substantially larger odds ratios - approximately 2.24–2.32 at 80% power and 2.54–2.69 at 90% power - would be detectable. Thus, the current study has limited power to detect the observed OR of 1.35.

5. Required for BIS PhD students, optional for MS, MPH, PhD students in other departments, and other students: Derive all 3 asymptotic test statistics – a) the Wald, b) the score and c) the likelihood ratio tests – to compare binary outcomes in two groups based upon a Bernoulli likelihood with the log link function. The log link function means that the Bernouilli parameter, p, has form $p = E(Y|X_{ij}) = exp\{\beta_0 + \beta^* X_{ij}\}$.

Solution:

Let $i = 1, 2$ indexes the two group, and $j = 1, ..., n$ indexes the individuals. We denote the independent observations as $\boldsymbol{Y} = (Y_1, \ldots, Y_n)^T$, where

$$f(Y_j \mid p_i) = p_i^{Y_j}(1 - p_i)^{1-Y_j}, \qquad Y_j \in \{0, 1\}.$$

For covariate X_{ij}, assume $X_{1j} = 0$ for $j = 1, \ldots, n_1$ and $X_{2j} = 1$ for $j = n_1 + 1, \ldots, n_1 + n_2$, where $n = n_1 + n_2$.

Using Bernoulli regression with the log link, we have

$$p_i = \exp(\beta_0 + \beta^* X_{ij}).$$

Let p_1 and p_2 denote the probabilities in the two groups:

$$p_1 = e^{\beta_0}, \qquad p_2 = e^{\beta_0 + \beta^*}.$$

For probabilities $0 < p_1, p_2 < 1$, the parameters must satisfy $\beta_0 < 0$ and $\beta_0 + \beta^* < 0$.

Let $\boldsymbol{\beta} = (\beta_0, \beta^*)^T$. Since

$$\beta^* = \log\left(\frac{p_2}{p_1}\right),$$

the hypothesis test is

$$H_0 : \beta^* = 0 \quad \text{versus} \quad H_A : \beta^* \neq 0,$$

which is equivalent to testing $H_0 : p_1 = p_2$ versus $H_A : p_1 \neq p_2$.

Define the numbers of successes in the two groups as

$$s_1 = \sum_{j=1}^{n_1} Y_j, \qquad s_2 = \sum_{j=n_1+1}^{n} Y_j.$$

Then the log-likelihood function of the observed data is

$$\begin{aligned}
\ell(\boldsymbol{\beta}) &= \sum_{j=1}^{n} \{Y_j \log p_i + (1 - Y_j) \log(1 - p_i)\} \\
&= \sum_{j=1}^{n} \left\{ Y_j(\beta_0 + \beta^* X_{ij}) + (1 - Y_j) \log(1 - e^{\beta_0 + \beta^* X_{ij}}) \right\} \\
&= \sum_{j=1}^{n_1} \left\{ Y_j \beta_0 + (1 - Y_j) \log(1 - e^{\beta_0}) \right\} \\
&\quad + \sum_{j=n_1+1}^{n} \left\{ Y_j(\beta_0 + \beta^*) + (1 - Y_j) \log(1 - e^{\beta_0 + \beta^*}) \right\} \\
&= s_1 \beta_0 + (n_1 - s_1) \log(1 - e^{\beta_0}) \\
&\quad + s_2(\beta_0 + \beta^*) + (n_2 - s_2) \log(1 - e^{\beta_0 + \beta^*}).
\end{aligned}$$

The following derivation assumes $0 < s_i < n_i$ for $i = 1, 2$. We take the first derivative of $\ell(\boldsymbol{\beta})$ w.r.t. $\boldsymbol{\beta}$:

$$\begin{aligned}
\boldsymbol{U}(\boldsymbol{\beta}) &= \frac{\partial \ell(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \\
&= \begin{pmatrix} \dfrac{\sum_{j=1}^{n_1} Y_j - n_1 p_1}{1 - p_1} + \dfrac{\sum_{j=n_1+1}^{n} Y_j - n_2 p_2}{1 - p_2} \\ \dfrac{\sum_{j=n_1+1}^{n} Y_j - n_2 p_2}{1 - p_2} \end{pmatrix} \\
&= \begin{pmatrix} \dfrac{s_1 - n_1 p_1}{1 - p_1} + \dfrac{s_2 - n_2 p_2}{1 - p_2} \\ \dfrac{s_2 - n_2 p_2}{1 - p_2} \end{pmatrix}.
\end{aligned}$$

where $p_1 = e^{\beta_0}$, $p_2 = e^{\beta_0 + \beta^*}$, $s_1 = \sum_{j=1}^{n_1} Y_j$, and $s_2 = \sum_{j=n_1+1}^{n} Y_j$. We solve $\boldsymbol{U}(\boldsymbol{\beta}) = \mathbf{0}$ to

obtain $\widehat{\boldsymbol{\beta}}$. Setting the second component equal to zero gives

$$s_2 - n_2\widehat{p}_2 = 0 \quad \Longrightarrow \quad \widehat{p}_2 = \frac{s_2}{n_2}.$$

Substituting this into the first component and setting it equal to zero gives

$$s_1 - n_1\widehat{p}_1 = 0 \quad \Longrightarrow \quad \widehat{p}_1 = \frac{s_1}{n_1}.$$

Thus,

$$\widehat{p}_1 = \frac{1}{n_1}\sum_{j=1}^{n_1} Y_j, \qquad \widehat{p}_2 = \frac{1}{n_2}\sum_{j=n_1+1}^{n} Y_j,$$

and the maximum likelihood estimator is

$$\boxed{\widehat{\boldsymbol{\beta}} = (\hat{\beta}_0, \hat{\beta}^*)^T = \begin{pmatrix} \log \widehat{p}_1 \\ \\ \log \widehat{p}_2 - \log \widehat{p}_1 \end{pmatrix}.}$$

The second derivatives are

$$\begin{aligned} \frac{\partial^2 \ell}{\partial \beta_0^2} &= -\frac{(n_1 - s_1)p_1}{(1-p_1)^2} - \frac{(n_2 - s_2)p_2}{(1-p_2)^2}, \\ \frac{\partial^2 \ell}{\partial \beta_0 \partial \beta^*} &= -\frac{(n_2 - s_2)p_2}{(1-p_2)^2}, \\ \frac{\partial^2 \ell}{\partial (\beta^*)^2} &= -\frac{(n_2 - s_2)p_2}{(1-p_2)^2} \end{aligned}$$

Since $s_1 = \sum_{j=1}^{n_1} Y_j$ is defined for group with $X_{1j} = 0$ and $s_2 = \sum_{j=n_1+1}^{n} Y_j$ is defined for group with $X_{2j} = 1$, then

$$\begin{aligned} E_{Y|X}[s_1] &= \sum_{j=1}^{n_1} E[Y_j | X_{1j} = 0] = \sum_{j=1}^{n_1} e^{\beta_0} = n_1 p_1; \\ E_{Y|X}[s_2] &= \sum_{j=n_1+1}^{n} E[Y_j | X_{1j} = 1] = \sum_{j=n_1+1}^{n} e^{\beta_0 + \beta^*} = n_2 p_2 \end{aligned}$$

Take the negative expectation for each second derivative

$$\begin{aligned}
-\mathbb{E}_{Y|X}\left[\frac{\partial^2 \ell}{\partial \beta_0^2}\right] &= \frac{(n_1 - E_{Y|X}[s_1])p_1}{(1-p_1)^2} + \frac{(n_2 - E_{Y|X}[s_2])p_2}{(1-p_2)^2} \\
&= \frac{(n_1 - n_1 p_1)p_1}{(1-p_1)^2} + \frac{(n_2 - n_2 p_2)p_2}{(1-p_2)^2} \\
&= \frac{n_1 p_1}{1-p_1} + \frac{n_2 p_2}{1-p_2}, \\
-\mathbb{E}_{Y|X}\left[\frac{\partial^2 \ell}{\partial \beta_0 \partial \beta^*}\right] &= \frac{(n_2 - E_{Y|X}[s_2])p_2}{(1-p_2)^2} \\
&= \frac{(n_2 - n_2 p_2)p_2}{(1-p_2)^2} \\
&= \frac{n_2 p_2}{1-p_2}, \\
-\mathbb{E}_{Y|X}\left[\frac{\partial^2 \ell}{\partial (\beta^*)^2}\right] &= \frac{(n_2 - E_{Y|X}[s_2])p_2}{(1-p_2)^2} \\
&= \frac{(n_2 - n_2 p_2)p_2}{(1-p_2)^2} \\
&= \frac{n_2 p_2}{1-p_2}.
\end{aligned}$$

The Fisher information matrix is

$$\begin{aligned}
\boldsymbol{I}(\boldsymbol{\beta}) &= -E_{\boldsymbol{\beta}}\left[\frac{\partial^2 \ell(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}\, \partial \boldsymbol{\beta}^T}\right] \\
&= \begin{pmatrix} \dfrac{n_1 p_1}{1-p_1} + \dfrac{n_2 p_2}{1-p_2} & \dfrac{n_2 p_2}{1-p_2} \\ \dfrac{n_2 p_2}{1-p_2} & \dfrac{n_2 p_2}{1-p_2} \end{pmatrix},
\end{aligned}$$

Using the formula for the inverse of a 2×2 matrix, the covariance matrix for $\hat{\beta}$ is

$$
\begin{aligned}
\mathrm{Var}(\hat{\boldsymbol{\beta}}) &= [\boldsymbol{I}(\boldsymbol{\beta})]^{-1} \\
&= \frac{1}{(\frac{n_1 p_1}{1-p_1} + \frac{n_2 p_2}{1-p_2})(\frac{n_2 p_2}{1-p_2}) - (\frac{n_2 p_2}{1-p_2})^2} \begin{pmatrix} \frac{n_2 p_2}{1-p_2} & -\frac{n_2 p_2}{1-p_2} \\ -\frac{n_2 p_2}{1-p_2} & \frac{n_1 p_1}{1-p_1} + \frac{n_2 p_2}{1-p_2} \end{pmatrix} \\
&= \begin{pmatrix} \frac{1-p_1}{n_1 p_1} & -\frac{1-p_1}{n_1 p_1} \\ -\frac{1-p_1}{n_1 p_1} & \frac{1-p_1}{n_1 p_1} + \frac{1-p_2}{n_2 p_2} \end{pmatrix}. \qquad (10)
\end{aligned}
$$

For hypothesis test

$$H_0 : \beta^* = 0 \quad \text{versus} \quad H_A : \beta^* \neq 0,$$

- **Wald Test:**

  The maximum likelihood estimator for β^* is

  $$\widehat{\beta}^* = \log \widehat{p}_2 - \log \widehat{p}_1$$

  The estimated variance for $\widehat{\boldsymbol{\beta}}$ is

  $$\widehat{\mathrm{Var}}(\widehat{\beta}^*) = \frac{1-\widehat{p}_1}{n_1 \widehat{p}_1} + \frac{1-\widehat{p}_2}{n_2 \widehat{p}_2}.$$

  Thus, the Wald test statistic is

  $$\boxed{Z_w^2 = \frac{(\widehat{\beta}^*)^2}{\widehat{\mathrm{Var}}(\widehat{\beta}^*)} = \frac{\{\log(\widehat{p}_2/\widehat{p}_1)\}^2}{\frac{1-\widehat{p}_1}{n_1 \widehat{p}_1} + \frac{1-\widehat{p}_2}{n_2 \widehat{p}_2}}.}$$

Equivalently, in terms of the success counts,

$$Z_w^2 = \frac{\left[\log\left\{\frac{s_2/n_2}{s_1/n_1}\right\}\right]^2}{\left(\frac{1}{s_1} - \frac{1}{n_1}\right) + \left(\frac{1}{s_2} - \frac{1}{n_2}\right)}.$$

Under H_0,

$$Z_w^2 \xrightarrow{d} \chi_1^2.$$

- **Score Test:**

  Under H_0, $\beta^* = 0$ and both groups have the common probability $p = e^{\beta_0}$. Then the log-likelihood function is

$$\begin{aligned}
\ell(\boldsymbol{\beta})|_{\boldsymbol{\beta^*}=0} &= \sum_{j=1}^{n_1} \left\{Y_j\beta_0 + (1-Y_j)\log(1-e^{\beta_0})\right\} \\
&\quad + \sum_{j=n_1+1}^{n} \left\{Y_j\beta_0 + (1-Y_j)\log(1-e^{\beta_0})\right\} \\
&= \sum_{j=1}^{n} \left\{Y_j\beta_0 + (1-Y_j)\log(1-e^{\beta_0})\right\} \\
&= (s_1+s_2)\beta_0 + (n-s_1-s_2)\log(1-e^{\beta_0}).
\end{aligned}$$

$$\left.\frac{\partial \ell(\boldsymbol{\beta})}{\partial \boldsymbol{\beta_0}}\right|_{\boldsymbol{\beta^*}=0} = (s_1+s_2) - (n-s_1-s_2)\frac{e^{\beta_0}}{1-e^{\beta_0}}$$

Set $\left.\frac{\partial \ell(\boldsymbol{\beta})}{\partial \boldsymbol{\beta_0}}\right|_{\boldsymbol{\beta^*}=0} = 0$ we have

$$\begin{aligned}
\frac{e^{\widetilde{\beta}_0}}{1-e^{\widetilde{\beta}_0}} &= \frac{s_1+s_2}{n-s_1-s_2} \\
e^{\widetilde{\beta}_0} &= \frac{s_1+s_2}{n}
\end{aligned}$$

The score function is

$$\boldsymbol{U}(\boldsymbol{\beta}) = \left.\frac{\partial \ell(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}\right|_{\boldsymbol{\beta}^*=0} = \begin{pmatrix} \dfrac{s_1 - n_1 p}{1-p} + \dfrac{s_2 - n_2 p}{1-p} \\ \dfrac{s_2 - n_2 p}{1-p} \end{pmatrix} = \begin{pmatrix} \dfrac{s_1 + s_2 - np}{1-p} \\ \dfrac{s_2 - n_2 p}{1-p} \end{pmatrix}.$$

Let $\widetilde{p} = e^{\widetilde{\beta}_0} = \frac{s_1+s_2}{n}$, and the score function evaluated at $\widetilde{\beta}_0$ is

$$\boldsymbol{U}(\widetilde{\boldsymbol{\beta}}) = \begin{pmatrix} \dfrac{s_1 + s_2 - n\widetilde{p}}{1-\widetilde{p}} \\ \dfrac{s_2 - n_2 \widetilde{p}}{1-\widetilde{p}} \end{pmatrix} = \begin{pmatrix} 0 \\ \dfrac{s_2 - n_2 \widetilde{p}}{1-\widetilde{p}} \end{pmatrix}.$$

By equation (10), the inverse of the Fisher information matrix under H_0 is

$$\left[\boldsymbol{I}(\boldsymbol{\beta})\right]^{-1}\Big|_{\boldsymbol{\beta}^*=0} = \begin{pmatrix} \dfrac{1-p}{n_1 p} & -\dfrac{1-p}{n_1 p} \\ -\dfrac{1-p}{n_1 p} & \dfrac{1-p}{n_1 p} + \dfrac{1-p}{n_2 p} \end{pmatrix}.$$

Therefore, the score test statistic evaluated at $\widetilde{\beta}_0$ is

$$\begin{aligned} \boldsymbol{Q} &= \boldsymbol{U}^T(\widetilde{\boldsymbol{\beta}})\boldsymbol{I}^{-1}(\widetilde{\boldsymbol{\beta}})\boldsymbol{U}(\widetilde{\boldsymbol{\beta}}) \\ &= \begin{pmatrix} 0 \\ \dfrac{s_2 - n_2\widetilde{p}}{1-\widetilde{p}} \end{pmatrix}^T \begin{pmatrix} \dfrac{1-\widetilde{p}}{n_1\widetilde{p}} & -\dfrac{1-\widetilde{p}}{n_1\widetilde{p}} \\ -\dfrac{1-\widetilde{p}}{n_1\widetilde{p}} & \dfrac{1-\widetilde{p}}{n_1\widetilde{p}} + \dfrac{1-\widetilde{p}}{n_2\widetilde{p}} \end{pmatrix} \begin{pmatrix} 0 \\ \dfrac{s_2 - n_2\widetilde{p}}{1-\widetilde{p}} \end{pmatrix} \\ &= \frac{n(1-\widetilde{p})}{n_1 n_2 \widetilde{p}} \left(\frac{s_2 - n_2\widetilde{p}}{1-\widetilde{p}}\right)^2 \\ &= \frac{n(s_2 - n_2\widetilde{p})^2}{n_1 n_2 \widetilde{p}(1-\widetilde{p})}. \end{aligned}$$

Note that $\hat{p}_1 = \frac{s_1}{n_1}$ and $\hat{p}_2 = \frac{s_2}{n_2}$, then $\widetilde{p} = \frac{s_1+s_2}{n} = \frac{n_1\widehat{p}_1+n_2\widehat{p}_2}{n}$, and we have:

$$\begin{aligned} s_2 - n_2\widetilde{p} &= n_2\widehat{p}_2 - n_2\frac{n_1\widehat{p}_1 + n_2\widehat{p}_2}{n} \\ &= \frac{n_1 n_2}{n}(\widehat{p}_2 - \widehat{p}_1). \end{aligned}$$

Substitute $s_2 - n_2\widetilde{p}$ with $\frac{n_1 n_2}{n}(\widehat{p}_2 - \widehat{p}_1)$,

$$\begin{aligned} \boldsymbol{Q} &= \frac{n(s_2 - n_2\widetilde{p})^2}{n_1 n_2 \widetilde{p}(1-\widetilde{p})} \\ &= \frac{n_1 n_2(\widehat{p}_2 - \widehat{p}_1)^2}{n\widetilde{p}(1-\widetilde{p})} \\ &= \frac{(\widehat{p}_2 - \widehat{p}_1)^2}{\widetilde{p}(1-\widetilde{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} \end{aligned}$$

Hence,

$$\boxed{\boldsymbol{Q} = \frac{(\widehat{p}_2 - \widehat{p}_1)^2}{\widetilde{p}(1-\widetilde{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Under H_0,

$$\boldsymbol{Q} \xrightarrow{d} \chi^2_1.$$

- **LRT:**

The log-likelihood in the unrestricted model is

$$\begin{aligned} \ell(\boldsymbol{\beta}) = s_1 \log p_1 + (n_1 - s_1)\log(1-p_1) \\ + s_2 \log p_2 + (n_2 - s_2)\log(1-p_2), \end{aligned}$$

where $p_1 = e^{\beta_0}$ and $p_2 = e^{\beta_0+\beta^*}$. Its maximized value is

$$\ell(\widehat{\boldsymbol{\beta}}) = s_1 \log \widehat{p}_1 + (n_1 - s_1)\log(1-\widehat{p}_1) + s_2 \log \widehat{p}_2 + (n_2 - s_2)\log(1-\widehat{p}_2) \tag{11}$$

Under H_0, the groups share a common probability, estimated by $\widetilde{p}$. Thus, the maximized log-likelihood in restricted model is

$$\ell(\widetilde{\boldsymbol{\beta}}) = (s_1 + s_2)\log\widetilde{p} + (n_1 + n_2 - s_1 - s_2)\log(1-\widetilde{p}). \tag{12}$$

The likelihood ratio is

$$\Lambda = \frac{L(\widetilde{\boldsymbol{\beta}})}{L(\widehat{\boldsymbol{\beta}})},$$

and the LRT statistic is

$$\begin{aligned} Z^2_{LRT} &= -2\log\Lambda \\ &= 2\left\{\ell(\widehat{\boldsymbol{\beta}}) - \ell(\widetilde{\boldsymbol{\beta}})\right\}. \end{aligned}$$

Plug in equation (11) and (12) gives

$$\begin{aligned} Z^2_{LRT} &= 2\left\{s_1\log\frac{\widehat{p}_1}{\widetilde{p}} + (n_1 - s_1)\log(\frac{1-\widehat{p}_1}{1-\widetilde{p}}) + s_2\log\frac{\widehat{p}_2}{\widetilde{p}} + (n_2 - s_2)\log(\frac{1-\widehat{p}_2}{1-\widetilde{p}})\right\} \\ &= 2\sum_{i=1}^{2}\left\{s_i\log\left(\frac{\widehat{p}_i}{\widetilde{p}}\right) + (n_i - s_i)\log\left(\frac{1-\widehat{p}_i}{1-\widetilde{p}}\right)\right\}. \end{aligned}$$

Hence,

$$\boxed{Z^2_{LRT} = 2\sum_{i=1}^{2}\left\{s_i\log\left(\frac{\widehat{p}_i}{\widetilde{p}}\right) + (n_i - s_i)\log\left(\frac{1-\widehat{p}_i}{1-\widetilde{p}}\right)\right\}.}$$

Under H_0,

$$Z^2_{LRT} \xrightarrow{d} \chi^2_1.$$

A R code

```r
# Packages
library(rootSolve)

# Load data: edit this path as needed
# load("path/to/MaxArt_HW1_2026.RData")

# Retain complete observations on Y and X
dat <- MaxArt_HW1_2026[
  complete.cases(MaxArt_HW1_2026[, c("Y", "X")]),
  c("Y", "X")
]

# Fit null and alternative models
fit0 <- glm(
  Y ~ 1,
  family = binomial(link = "logit"),
  data = dat
)

fit1 <- glm(
  Y ~ X,
  family = binomial(link = "logit"),
  data = dat
)

# Extract design and model quantities
n <- nrow(dat)
p_x <- mean(dat$X)
b00 <- unname(coef(fit0)[1])
b0a <- unname(coef(fit1)[1])
b1_hat <- unname(coef(fit1)["X"])
```

```r
or_hat <- exp(b1_hat)
p0_hat <- plogis(b0a)
p1_hat <- plogis(b0a + b1_hat)

design_summary <- data.frame(
  Parameter = c(
    "Sample size",
    "P(X = 1)",
    "beta_0 under H0",
    "beta_0 under HA",
    "beta_1 under HA",
    "Observed odds ratio",
    "P(Y = 1 | X = 0)",
    "P(Y = 1 | X = 1)"
  ),
  Estimate = c(
    n, p_x, b00, b0a, b1_hat,
    or_hat, p0_hat, p1_hat
  )
)

print(design_summary, row.names = FALSE)

# ------------------------------------------------------------------------------
# Question 3: Wald, score, and likelihood-ratio tests
# ------------------------------------------------------------------------------

alpha <- 0.05
critical_value <- qchisq(1 - alpha, df = 1)

# Wald test
wald_se <- sqrt(vcov(fit1)["X", "X"])
wald_z <- b1_hat / wald_se
```

```r
wald_stat <- unname(wald_z^2)
wald_p <- pchisq(wald_stat, df = 1, lower.tail = FALSE)

# Score test adjusted for the nuisance intercept
p_null <- plogis(b00)
U_beta <- sum(dat$X * (dat$Y - p_null))
I_beta_efficient <- p_null * (1 - p_null) *
  sum((dat$X - mean(dat$X))^2)
score_stat <- unname(U_beta^2 / I_beta_efficient)
score_p <- pchisq(score_stat, df = 1, lower.tail = FALSE)

# Likelihood-ratio test
lrt_stat <- unname(deviance(fit0) - deviance(fit1))
lrt_p <- pchisq(lrt_stat, df = 1, lower.tail = FALSE)

q3_results <- data.frame(
  Test = c("Wald", "Score", "Likelihood ratio"),
  Statistic = c(wald_stat, score_stat, lrt_stat),
  df = c(1, 1, 1),
  P_value = c(wald_p, score_p, lrt_p)
)

q3_results$Decision <- ifelse(
  q3_results$P_value < alpha,
  "Reject H0",
  "Do not reject H0"
)

print(
  transform(
    q3_results,
    Statistic = round(Statistic, 4),
    P_value = round(P_value, 4)
```

```
  ),
  row.names = FALSE
)

# ------------------------------------------------------------------------------
# Question 4: target noncentrality parameters
# ------------------------------------------------------------------------------

lambda_80 <- 7.85
lambda_90 <- 10.50

# Wald NCP
wald_ncp <- function(beta, sample_size, p_x, b0) {
  Ib1 <-
    (1 / (1 - p_x)) *
    ((1 + exp(b0))^2 / exp(b0)) +
    (1 / p_x) *
    ((1 + exp(b0 + beta))^2 / exp(b0 + beta))

  sample_size * beta^2 / Ib1
}

lambda_w <- wald_ncp(
  beta = b1_hat,
  sample_size = n,
  p_x = p_x,
  b0 = b0a
)

power_wald <- 1 - pchisq(
  critical_value,
  df = 1,
  ncp = lambda_w
```

```r
)

Ib1_observed <-
  (1 / (1 - p_x)) *
  ((1 + exp(b0a))^2 / exp(b0a)) +
  (1 / p_x) *
  ((1 + exp(b0a + b1_hat))^2 /
     exp(b0a + b1_hat))

n_wald_80 <- ceiling(
  lambda_80 / (b1_hat^2 / Ib1_observed)
)
n_wald_90 <- ceiling(
  lambda_90 / (b1_hat^2 / Ib1_observed)
)

# Score NCP
score_ncp <- function(beta, sample_size, p_x, b0a) {
  p_0 <- plogis(b0a)
  p_1 <- plogis(b0a + beta)

  sample_size *
    (p_x * (1 - p_x) * (p_1 - p_0)^2) /
    (
      p_x * p_0 * (1 - p_0) +
      (1 - p_x) * p_1 * (1 - p_1)
    )
}

lambda_s <- score_ncp(
  beta = b1_hat,
  sample_size = n,
  p_x = p_x,
```

```
  b0a = b0a
)

power_score <- 1 - pchisq(
  critical_value,
  df = 1,
  ncp = lambda_s
)

p_0_observed <- plogis(b0a)
p_1_observed <- plogis(b0a + b1_hat)

score_rate_observed <-
  (p_x * (1 - p_x) *
     (p_1_observed - p_0_observed)^2) /
  (
    p_x * p_0_observed * (1 - p_0_observed) +
    (1 - p_x) * p_1_observed * (1 - p_1_observed)
  )

n_score_80 <- ceiling(lambda_80 / score_rate_observed)
n_score_90 <- ceiling(lambda_90 / score_rate_observed)

# Likelihood-ratio NCP components
lrt_components <- function(beta, p_x, b0a) {
  p0 <- plogis(b0a)
  p1 <- plogis(b0a + beta)
  p_star <- (1 - p_x) * p0 + p_x * p1
  b_tilde <- log(p_star / (1 - p_star))

  D_star <- 2 * (
    p_x * (
      p1 * (b0a + beta) -
```

```
      p_star * b_tilde -
      log(
        (1 + exp(b0a + beta)) /
        (1 + exp(b_tilde))
      )
    ) +
    (1 - p_x) * (
      p0 * b0a -
      p_star * b_tilde -
      log(
        (1 + exp(b0a)) /
        (1 + exp(b_tilde))
      )
    )
  )

  M_t <-
    (
      p_x * p1 * (1 - p1) +
      (1 - p_x) * p0 * (1 - p0)
    ) /
    (p_star * (1 - p_star))

  list(
    D_star = D_star,
    M_t = M_t,
    p_star = p_star,
    b_tilde = b_tilde
  )
}

lrt_observed <- lrt_components(
  beta = b1_hat,
```

```r
  p_x = p_x,
  b0a = b0a
)

lambda_lrt <-
  n * lrt_observed$D_star +
  1 - lrt_observed$M_t

power_lrt <- 1 - pchisq(
  critical_value,
  df = 1,
  ncp = lambda_lrt
)

n_lrt_80 <- ceiling(
  (lambda_80 - 1 + lrt_observed$M_t) /
    lrt_observed$D_star
)
n_lrt_90 <- ceiling(
  (lambda_90 - 1 + lrt_observed$M_t) /
    lrt_observed$D_star
)

# ------------------------------------------------------------------------------
# MDE calculations using the rootSolve approach
# ------------------------------------------------------------------------------

mde_start <- 0.8

# Wald MDE equation
power.eql.wald <- function(b1, para) {
  p_x <- para[1]
  b0 <- para[2]
```

```r
  n <- para[3]
  target_lambda <- para[4]

  Ib1 <-
    (1 / (1 - p_x)) *
    ((1 + exp(b0))^2 / exp(b0)) +
    (1 / p_x) *
    ((1 + exp(b0 + b1))^2 / exp(b0 + b1))

  n * b1^2 / Ib1 - target_lambda
}

find.wald.80 <- multiroot(
  f = power.eql.wald,
  start = mde_start,
  para = c(p_x, b0a, n, lambda_80),
  positive = TRUE
)
b1_wald_mde_80 <- unname(find.wald.80$root)
OR_wald_mde_80 <- exp(b1_wald_mde_80)

find.wald.90 <- multiroot(
  f = power.eql.wald,
  start = b1_wald_mde_80,
  para = c(p_x, b0a, n, lambda_90),
  positive = TRUE
)
b1_wald_mde_90 <- unname(find.wald.90$root)
OR_wald_mde_90 <- exp(b1_wald_mde_90)

# Score MDE equation
power.eql.score <- function(b1, para) {
  b0a <- para[1]
```

```r
  n <- para[2]
  p_x <- para[3]
  target_lambda <- para[4]

  p_y0 <- plogis(b0a)
  p_y1 <- plogis(b0a + b1)

  n *
    (p_x * (1 - p_x) * (p_y1 - p_y0)^2) /
    (
      p_x * p_y0 * (1 - p_y0) +
      (1 - p_x) * p_y1 * (1 - p_y1)
    ) -
    target_lambda
}

find.score.80 <- multiroot(
  f = power.eql.score,
  start = mde_start,
  para = c(b0a, n, p_x, lambda_80),
  positive = TRUE
)
b1_score_mde_80 <- unname(find.score.80$root)
OR_score_mde_80 <- exp(b1_score_mde_80)

find.score.90 <- multiroot(
  f = power.eql.score,
  start = b1_score_mde_80,
  para = c(b0a, n, p_x, lambda_90),
  positive = TRUE
)
b1_score_mde_90 <- unname(find.score.90$root)
OR_score_mde_90 <- exp(b1_score_mde_90)
```

```r

# Likelihood-ratio MDE equation
power.eql.lrt <- function(b1, para) {
  b0a <- para[1]
  n <- para[2]
  p_x <- para[3]
  target_lambda <- para[4]

  z <- lrt_components(
    beta = b1,
    p_x = p_x,
    b0a = b0a
  )

  n * z$D_star + 1 - z$M_t - target_lambda
}

find.lrt.80 <- multiroot(
  f = power.eql.lrt,
  start = mde_start,
  para = c(b0a, n, p_x, lambda_80),
  positive = TRUE
)
b1_lrt_mde_80 <- unname(find.lrt.80$root)
OR_lrt_mde_80 <- exp(b1_lrt_mde_80)

find.lrt.90 <- multiroot(
  f = power.eql.lrt,
  start = b1_lrt_mde_80,
  para = c(b0a, n, p_x, lambda_90),
  positive = TRUE
)
b1_lrt_mde_90 <- unname(find.lrt.90$root)
```

```r
OR_lrt_mde_90 <- exp(b1_lrt_mde_90)

# Concise result table
q4_complete_table <- data.frame(
  Test = rep(
    c("Wald", "Score", "Likelihood ratio"),
    each = 2
  ),
  Target_power = rep(c(0.80, 0.90), times = 3),
  Power_at_current_n = rep(
    c(power_wald, power_score, power_lrt),
    each = 2
  ),
  Required_sample_size = c(
    n_wald_80, n_wald_90,
    n_score_80, n_score_90,
    n_lrt_80, n_lrt_90
  ),
  MDE_log_OR = c(
    b1_wald_mde_80, b1_wald_mde_90,
    b1_score_mde_80, b1_score_mde_90,
    b1_lrt_mde_80, b1_lrt_mde_90
  ),
  MDE_OR = c(
    OR_wald_mde_80, OR_wald_mde_90,
    OR_score_mde_80, OR_score_mde_90,
    OR_lrt_mde_80, OR_lrt_mde_90
  )
)

print(
  transform(
    q4_complete_table,
```

```
    Power_at_current_n = round(Power_at_current_n, 4),
    MDE_log_OR = round(MDE_log_OR, 4),
    MDE_OR = round(MDE_OR, 4)
  ),
  row.names = FALSE
)
```